

BUSINESS REQUIREMENTS SPECIFICATION

Supply Chain Exception Control System

Explainable purchase-order delay and stockout exception control

CONTROL PRINCIPLE Decision support, not autonomous control. Every exception must be explainable; every human action, approval, override, and resolution must be traceable.

Document field	Value
Document status	Version 1.0 - proposed BRS for approval; governed by the approved Project Charter v1.0
Prepared for	Portfolio project - simulated Australian mid-sized distributor/manufacturer
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Date	19 July 2026
Primary operational grain	One purchase-order line risk episode
Core release boundary	Data foundation, deterministic risk engine, operational exception workflow, audit trail, and Power BI
Primary platform	PostgreSQL deployed; SQLite permitted only for local development/testing

Important evidence statement

All volumes, baselines, service levels, performance thresholds, and benefits in this document are proposed portfolio acceptance targets or modelling assumptions. They are not achieved business results.

Document control

Item	Definition
Governing source	Supply Chain Exception Control System Project Charter, Version 1.0, dated 19 July 2026.
Baseline precedence	The confirmed decisions in the commissioning instruction override conflicting charter recommendations. Conflicts are recorded in Appendix C.
Requirement language	Must = mandatory for the stated phase; Should = important but negotiable; Could = optional if capacity remains; Will Not = explicitly excluded from the stated release.
Phase labels	P0 Design; P1 Data foundation; P2 Risk engine; P3 Operational MVP; P4 Power BI/core release; P5 Automation optional; P6 Controlled AI optional; Future = outside current portfolio release.
Acceptance status	A requirement is accepted only when its stated criterion is met and referenced evidence is retained. Failed targets must be disclosed.
Change control	Changes after approval require a decision-log entry, impact assessment, requirement version update, and retest of affected controls.

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1. Document purpose

This specification translates the approved Project Charter v1.0 into specific, testable business, functional, control, data, reporting, and governance requirements. It establishes what the portfolio system must demonstrate before data modelling and implementation begin. It does not prescribe database SQL, Python code, Power BI measures, or user-interface code.

SCOPE OF AUTHORITY The charter remains the authority for the business problem and project boundary. This BRS resolves the confirmed baseline decisions supplied with the commission and makes proposed design assumptions explicit; it does not convert assumptions into real-world evidence.

2. Business problem summary

The simulated organisation uses ERP extracts, inventory snapshots, demand data, supplier history, spreadsheets, email, and individual judgement to review approximately 1,500 open PO lines per operational day. The current process spreads attention across normal and risky orders, detects some threats late, applies inconsistent urgency, fragments ownership and action evidence, and makes management reconstruction difficult.

Control gap	Required response
Coverage	Validate and assess all eligible open PO lines in each successful run.
Timing	Identify material risk while expediting, substitution, allocation, or replanning remains possible.
Priority	Use one transparent, deterministic, versioned risk priority score and severity matrix.
Accountability	Make owner, due action, escalation state, and approval responsibility explicit.
Evidence	Preserve source-to-score-to-action-to-resolution-to-report lineage without destructive UI history edits.

3. Project objectives

ID	Objective	Proposed portfolio outcome
O1	Focus operational effort	Reduce assumed daily screening from 120 to <=35 active minutes while maintaining agreed risk coverage.
O2	Detect earlier	Surface high/critical simulated risks >=2 Melbourne business days before need date where an input signal exists.
O3	Standardise priority	Apply one documented rule scorecard and severity matrix to 100% of eligible lines.
O4	Create ownership	Assign >=95% of high/critical exceptions within 4 Melbourne business hours.
O5	Preserve control evidence	Capture 100% of controlled owner, priority, status, approval, override, and resolution changes with actor, time, before/after, and reason.
O6	Enable management learning	Refresh reconciled Power BI monitoring daily for backlog, ageing, SLA, outcomes, suppliers, interventions, overrides, and data health.

ID	Objective	Proposed portfolio outcome
O7	Govern AI safely	If built, ensure 100% of AI outputs are reviewed and 0 AI outputs autonomously mutate or execute operational decisions.

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
BR-001	Focused worklist. The system shall present a ranked exception worklist derived from eligible open PO lines. Rationale: Operational effort must be directed to the small subset requiring action.	Must P3	In a timed scripted daily scenario, the operator identifies and records the priority work in <=35 active minutes; the assumed baseline of 120 minutes is shown separately.	O1 / Manual screening time	Timed usability test UT-01.
BR-002	Early warning. High/critical simulated adverse outcomes shall be measured for useful lead time. Rationale: Earlier warning preserves response options.	Must P2	Median useful lead time is >=2 Melbourne business days and >=70% of correctly detected high/critical outcomes are first flagged >=2 days before need date where source signals existed.	O2 / Useful lead time	Backtest AT-02 on unseen scenarios.
BR-003	Standard assessment. Every eligible line in a successful run shall receive one versioned risk priority score. Rationale: Inconsistent individual screening is a priority-control gap.	Must P2	100% of eligible lines are scored; >=99.5% of eligible received lines are covered; exclusions carry an explicit reason.	O3 / Coverage	Coverage reconciliation AT-03.
BR-004	Accountability. High and critical exceptions shall have explicit ownership and due actions. Rationale: Unowned exceptions create silent risk.	Must P3	>=95% assigned within 4 Melbourne business hours; all active high/critical records show owner or breached-unassigned state.	O4 / Assignment SLA	SLA scenario WT-04.
BR-005	Traceable decisions. All controlled human and automated state changes shall be reconstructable. Rationale: Management and audit need evidence, not overwritten snapshots.	Must P3	100% of sampled controlled changes contain actor/system identity, timestamp, before/after, reason/source, and linked exception/run.	O5 / Audit completeness	Audit completeness AT-05.
BR-006	Management visibility. Management reporting shall reconcile to the operational store and disclose freshness/failure. Rationale: A separate management view supports learning without replacing the queue.	Must P4	Counts reconcile by run/date/severity/status/site/supplier; latest successful dataset age <=26 hours after daily refresh.	O6 / Dashboard freshness	BI reconciliation RT-01.
BR-007	Human control over AI. Optional AI shall remain advisory and unable to mutate or execute workflow decisions. Rationale: Plausible output must not become uncontrolled action.	Must P6	Across all AI test cases, 100% are reviewed and 0 create, close, reprioritise, reassign, approve, suppress, resolve, or execute actions.	O7 / AI human review	Adversarial and permission test AI-AT-01.
BR-008	Honest evidence. Portfolio outputs shall distinguish proposed targets, simulated test results, and any future real results. Rationale: Synthetic evidence cannot prove realised commercial benefit.	Must P0-P6	Every displayed performance result is labelled proposed target, simulated measured result, or real-pilot result; no unsupported savings claim appears.	All objectives / Evidence integrity	Document and screen review GOV-T-08.

4. Stakeholder requirements and user stories

User stories describe the business outcome; the associated requirement and Given/When/Then criteria define acceptance. Simulated role switching is a demonstration mechanism only and is not production authentication.

ID	Role	User story	Priority / phase	Trace	Test
US-001	Buyer / procurement analyst	As a buyer or procurement analyst, I need a ranked queue of my assigned and unassigned exceptions, so that I can focus on the PO lines needing action now.	Must	P3	O1, O4 / screening and assignment
US-002	Buyer / procurement analyst	As a buyer or procurement analyst, I need to see score reasons, inventory exposure, supplier context, actions, and evidence in one exception record, so that I can investigate and document a defensible response.	Must	P3	O3, O5 / explainability and audit
US-003	Inventory planner	As an inventory planner, I need item-site projected balance, need-date impact, mitigation, and demand assumptions, so that I can assess stockout consequence and propose substitution, allocation, or replanning.	Must	P3	O2, O3 / useful lead time
US-004	Supply-chain manager	As a supply-chain manager, I need overdue, unassigned, critical, approval, and supplier-pattern views, so that I can intervene, approve material exceptions, and manage systemic risk.	Must	P3-P4	O4, O6 / SLA and backlog
US-005	Data / system administrator	As a data or system administrator, I need run status, validation failures, quarantined records, rule versions, and reconciliation results, so that I can release only trustworthy processing outputs.	Must	P1-P5	O3, O5 / validity and reliability
US-006	Governance / audit reviewer	As a governance or audit reviewer, I need a chronological, non-destructive history of automated and human decisions, so that I can reconstruct who changed what, why, under which rule version, and with which approval.	Must	P3	O5 / audit completeness
US-007	Supply-chain manager	As a supply-chain manager, I need reconciled Power BI measures for backlog, ageing, SLA, reasons, outcomes, suppliers, overrides, and data health, so that I can evaluate control performance and improvement priorities.	Must	P4	O6 / reporting KPIs

Key Given / When / Then acceptance criteria

Story	Given	When	Then
US-001	Given a successful run with assigned, unassigned, overdue, and multiple-severity exceptions	When the buyer opens the operational queue and applies owner/site/severity filters	Then the returned records match the store, default ordering places critical and breached items first, and no excluded record is lost from audit history.
US-002	Given an open exception with three triggered rules and previous actions	When the buyer opens the detail and records a supplier contact, commitment date, next action, note, and evidence reference	Then the total score equals displayed contributions and a new attributed action event appears without overwriting prior events.

Story	Given	When	Then
US-003	Given a line where projected available balance falls below zero before expected receipt	When the planner reviews item-site exposure and records an approved substitute that covers the need	Then the mitigation is visible, its score contribution is applied under the active rule version, and approval/evidence rules are enforced.
US-004	Given a severity-reduction override request on a high exception	When a manager approves it with reason and expiry	Then original and approved values remain visible, the approval is attributed, expiry is scheduled, and the override appears in reporting.
US-005	Given a run with a duplicate key and a missing mandatory promised/need date	When validation executes	Then the run is stopped if the defect is critical, affected records are quarantined, scoring output is not published, and the failure is visible.
US-006	Given an exception moved through assignment, investigation, action, monitoring, resolution, closure, and reopening	When the reviewer opens its history	Then every transition, actor/system identity, Melbourne timestamp, reason, evidence/approval reference, and before/after value is shown in sequence.
US-007	Given a successful run followed by a failed scheduled run	When the manager opens Power BI	Then the model continues to show the latest successful dataset, displays its age and the failed-run indicator, and reconciles totals to that successful run.

5. Functional requirements

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
FR-001	Source ingestion. Ingest PO header/line, receipt, inventory, demand, confirmation, item, supplier, site, and supplier-performance datasets into a dated run. Rationale: All required risk signals need controlled source context.	Must P1	A complete test package loads with source filename/type, extracted time, row count, and run ID recorded for every dataset.	O3, O5 / coverage, lineage	Integration test IT-01.
FR-002	Run identity. Create a unique immutable run ID and Melbourne start/end timestamps for each attempt. Rationale: Lineage and reconciliation depend on stable runs.	Must P1	30 attempts generate 30 unique IDs, including failed attempts; no ID is reused.	O5 / audit completeness	Uniqueness test IT-02.
FR-003	Validation gate. Execute mandatory validation before transformation and scoring. Rationale: Invalid critical data can create misleading low risk.	Must P1	Injected critical defects stop publication in 100% of cases; valid controls and counts are stored.	O3 / critical data validity	DQ-T-01.
FR-004	Quarantine. Retain non-critical rejected records with defect code, field, value, source, and run. Rationale: Defects must be visible and correctable.	Must P1	Each injected non-critical defect appears once in quarantine and is excluded/qualified according to its rule.	O5, O6 / data health	DQ-T-02.
FR-005	Eligibility. Determine eligible open PO lines using documented status, quantity, receipt, and date rules. Rationale: The scored population must be reproducible.	Must P2	Expected eligible, excluded, closed, cancelled, and fully received test lines match 100%; each exclusion has a code.	O3 / control coverage	Rule test RE-01.
FR-006	Risk priority score. Calculate a deterministic 0-100 risk priority score for every eligible line. Rationale: Priority must be transparent and reproducible.	Must P2	Same inputs, configuration, and version produce identical scores; all totals remain 0-100.	O3 / explainability	Determinism test RE-02.
FR-007	Contribution breakdown. Persist every evaluated rule with trigger result, observed value, threshold, points, reason, and availability. Rationale: Users must know why a score exists.	Must P2	100% of sampled exceptions reproduce total score from stored contributions and show unavailable signals distinctly.	O3 / explainability completeness	Recalculation test RE-03.
FR-008	Rule version. Attach one approved rule version and effective interval to every score. Rationale: Rule changes must not rewrite history.	Must P2	Scores before and after a version change retain their original versions; no retroactive alteration occurs.	O5 / audit completeness	Version test RE-04.
FR-009	Severity band. Derive Monitor/Medium/High/Critical from the active score matrix and persist the derived band. Rationale: Response standards rely on severity.	Must P2	Boundary tests at 0,29,30,49,50,69,70,100 return expected bands.	O3 / coverage	Boundary test RE-05.
FR-010	Candidate evaluation. Create a candidate-risk record for each eligible evaluation before lifecycle decisions. Rationale: Score evidence must exist even when no exception opens.	Must P2	Every eligible line/run has one candidate record linked to its score and disposition.	O3, O5 / coverage	Cardinality test RE-06.
FR-011	Persistent episode. Maintain one stable exception ID for a continuous PO-line risk episode across runs. Rationale: Repeated scores must not fragment work.	Must P3	A multi-run scenario updates one episode until closure; material post-closure deterioration produces a linked reopened episode/state per rule.	O4, O5 / backlog integrity	Lifecycle test LC-01.
FR-012	Deduplication. Prevent more than one active exception for the same documented episode key. Rationale: Duplicates recreate workload and distort reporting.	Must P3	Concurrent/repeated evaluation produces one active exception and linked evaluations; duplicate count is zero.	O1, O6 / backlog	Uniqueness test LC-02.

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
FR-013	Queue. Show active exceptions with severity, owner, site, supplier, item, dates, age, next action, SLA state, and score. Rationale: Operators need a complete workload.	Must P3	All required fields are present and match the store for a 100-record sample.	O1, O4 / screening	UI/store reconciliation WT-01.
FR-014	Queue filters. Filter by owner/unassigned, severity, status, site, supplier, buyer group, need date, stockout date, age, and overdue action. Rationale: Different roles require focused slices.	Must P3	Each filter and two-filter combination returns the exact expected set in scripted cases.	O1, O4 / screening	Filter test WT-05.
FR-015	Default ranking. Rank by breached/unassigned criticality, severity, need/stockout date, criticality, and time since action using a documented tie-breaker. Rationale: Queue order must be stable and defensible.	Must P3	A fixed 25-record fixture sorts exactly as the published precedence and stable tie-breaker specify.	O1, O3 / screening	Ordering test WT-06.
FR-016	Exception detail. Show source facts, projected exposure, score breakdown/version, owner/SLA, lifecycle, action/approval/evidence history, and outcome. Rationale: Investigation should avoid fragmented context.	Must P3	A reviewer locates each required element for all five reference scenarios and values reconcile.	O1, O3, O5	Scenario review WT-02.
FR-017	Assignment. Support configured automatic assignment and authorised manual assignment/reassignment. Rationale: Exceptions need explicit accountable ownership.	Must P3	Mapped cases auto-assign; unmapped cases enter unassigned queue; reassignment captures before/after, actor, reason, and time.	O4 / assignment SLA	Assignment test OW-01.
FR-018	Action capture. Record action type, note, supplier commitment, next-action date, evidence reference, actor, and time. Rationale: Fragmented action evidence is a control gap.	Must P3	Mandatory fields are enforced by action type; successful entry is append-only and visible chronologically.	O5 / audit completeness	Action test AC-01.
FR-019	Status transition control. Permit only documented lifecycle transitions and enforce required fields/authority. Rationale: Free-form statuses destroy process control.	Must P3	All allowed transitions pass; all prohibited transitions fail without changing state and are logged as rejected attempts where configured.	O5 / audit completeness	State-machine test LC-03.
FR-020	Override workflow. Create, approve/reject, expire, and report material overrides without changing the original score evidence. Rationale: Overrides can otherwise hide unresolved risk.	Must P3	Severity reduction, accepted residual risk, criticality override, and closure without objective evidence require reason, evidence, expiry where applicable, and manager decision.	O5 / override rate	Approval test AP-01.
FR-021	Resolution. Capture resolution code, cause code, evidence, residual risk, resolver, and time before Resolved. Rationale: Resolution must be structured and verifiable.	Must P3	All resolution paths reject missing mandatory evidence/approval; valid paths retain complete fields.	O5 / cycle time	Resolution test LC-04.
FR-022	Closure. Restrict Closed to verified Resolved exceptions and retain closure authority/evidence. Rationale: Closing work is distinct from risk resolution.	Must P3	Direct active-to-Closed attempts fail; valid Resolved-to-Closed succeeds only for authorised role and records evidence.	O5 / audit completeness	Closure test LC-05.
FR-023	Reopen. Automatically or manually reopen closed/resolved/suppressed risk under documented material-deterioration rules. Rationale: Risk can recur after apparent recovery.	Must P3	Reference scenarios reopen exactly once, link prior state/episode, reset applicable SLA, and retain prior closure.	O2, O5 / recall	Reopen test LC-06.
FR-024	Audit view. Display chronological automated and human event history with filters and export. Rationale: Reviewers need reconstructable evidence.	Must P3	The golden episode audit export matches the expected event ledger exactly.	O5 / audit completeness	Golden-record test AT-05.

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
FR-025	BI publication. Publish curated, run-identified data for Power BI only after successful gates. Rationale: Reporting must not consume partial/failed results.	Must P4	Failed runs publish no new current dataset; successful run exports reconcile row counts and checksums/control totals.	O6 / freshness	Publication test RT-01.
FR-026	Scheduled processing. Support scheduled validation, scoring, tests, and exports using GitHub Actions after local processing passes. Rationale: Automation demonstrates repeatability, not enterprise scheduling.	Should P5	30 simulated scheduled executions achieve >=98% success or failures are disclosed; every failure has visible status/log metadata.	O6 / batch reliability	Schedule test OPS-T-01.
FR-027	Notification sequence. Do not implement email notification until the in-app queue, SLA, lifecycle, and audit controls pass acceptance. Rationale: Notifications must not precede the system of record for action.	Must P3-P5	Release checklist shows queue controls accepted before any notification feature is enabled.	O4, O5	Gate review GATE-05.
FR-028	Local-first deployment. Complete and accept a local PostgreSQL/Streamlit/Power BI demonstration before any hosted demonstration. Rationale: Hosting adds security and persistence risk.	Must P4	Local core acceptance report is approved before a hosted deployment branch or public endpoint is enabled.	All core objectives	Release gate GATE-04.

6. Non-functional requirements

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
NFR-001	Batch performance. Validate and score at least 50,000 PO lines within 5 minutes in the documented test environment. Rationale: The daily process must be practicable.	Must P2	Five repeated runs meet <=5 minutes; hardware, software, row counts, and warm/cold conditions are recorded.	O3 / batch performance	Performance test PT-01.
NFR-002	Operational responsiveness. Return the default 1,500-line open-queue view and standard filters within 3 seconds locally. Rationale: Operators must not wait during routine triage.	Should P3	95th percentile response <=3 seconds across 30 scripted requests in the documented environment. Proposed target.	O1 / screening	Performance test PT-02.
NFR-003	Repeatability. Produce identical outputs for identical approved input, configuration, rule version, and random seed. Rationale: Portfolio evidence must be reproducible.	Must P1-P5	Hash/control totals match across three clean executions.	O3, O5	Repeatability test PT-03.
NFR-004	Atomic publication. Prevent users and Power BI from seeing a partially processed current run. Rationale: Partial outputs can mislead decisions.	Must P1-P4	Injected failure at each stage leaves the last successful current dataset unchanged.	O5, O6	Failure-injection test PT-04.
NFR-005	Batch reliability. Complete >=98% of 30 simulated scheduled runs or explicitly report shortfall. Rationale: Reliability is a proposed control target.	Should P5	Success ratio and causes are published for exactly 30 attempts; failures remain visible.	O6 / batch reliability	Reliability test OPS-T-01.
NFR-006	Time handling. Store and display SLA-relevant times consistently for Australia/Melbourne, including DST boundaries. Rationale: The project time zone is binding.	Must P1-P4	Autumn and spring DST test cases calculate and display the expected local time and business-duration result.	O4, O5 / SLA	Time test PT-06.
NFR-007	Capacity. Support the baseline of 120 suppliers, 1,000 SKUs, 50,000+ historical PO lines, 24 months, two sites, and 1,500 open lines/day. Rationale: The portfolio scale must match its approved context.	Must P1-P4	Reference dataset meets or exceeds every volume and all core tests complete without truncation.	All / scale	Capacity test PT-07.
NFR-008	Explainability availability. Make stored score contributions and rule version available even if the current rule configuration changes. Rationale: Historical decisions must remain interpretable.	Must P2-P4	All historical sampled exceptions render their original explanation after two simulated rule changes.	O3, O5	Historical explainability test PT-08.
NFR-009	Portability boundary. Use PostgreSQL as deployed source of truth; SQLite may be used only for local development/testing and shall not define incompatible business semantics. Rationale: The confirmed architecture supersedes the charter SQLite-first default.	Must P1-P4	The deployed demo runs on PostgreSQL; any SQLite test suite produces matching control totals for supported tests and is labelled non-deployed.	O5, O6	Cross-store conformance PT-09.

7. Data-quality requirements

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
DQ-001	Stable keys. Enforce stable PO line, item, supplier, site, receipt, and run identifiers. Rationale: Episodes and lineage cannot work with ambiguous keys.	Must P1	Zero duplicate primary keys in accepted data; every injected duplicate is stopped or quarantined per severity.	O3, O5 / validity	Key test DQ-T-01.
DQ-002	Referential integrity. Validate every accepted PO line and fact reference against required master data. Rationale: Orphan facts can distort risk.	Must P1	100% accepted references resolve; injected orphans are quarantined/stopped with code.	O3 / validity	Referential test DQ-T-02.
DQ-003	Quantity integrity. Validate non-negative ordered/received/open quantities and receipt not exceeding tolerance without an explicit exception. Rationale: Quantity defects change eligibility and exposure.	Must P1	Boundary fixture produces the expected pass/fail result for every quantity case.	O3 / validity	Boundary test DQ-T-03.
DQ-004	Date integrity. Validate order, request/need, promise/confirmation, expected receipt, and receipt chronology using documented exceptions. Rationale: Date signals drive urgency and delay.	Must P1	All valid/invalid chronology fixtures are classified exactly as specified.	O2, O3 / validity	Date test DQ-T-04.
DQ-005	Freshness. Measure and display source age for PO, receipt, inventory, demand, and supplier data. Rationale: Stale data can imply false safety.	Must P1-P4	Every run stores freshness; critical staleness stops/qualifies scoring; dashboard displays age and threshold.	O2, O6 / freshness	Freshness test DQ-T-05.
DQ-006	Mandatory completeness. Define mandatory fields by entity and scenario and calculate completeness. Rationale: Missing critical signals require explicit treatment.	Must P1	Accepted critical-field completeness >=99.5%; critical missingness invokes the configured gate.	O3 / critical validity	Completeness test DQ-T-06.
DQ-007	Missing-signal treatment. Mark insufficient/unavailable supplier or demand signals as unavailable, not zero risk. Rationale: Defaulting missing data to safe creates false confidence.	Must P2	All missing-signal fixtures display unavailable/qualified state and never receive an implicit zero-risk contribution.	O2, O3	Rule test DQ-T-07.
DQ-008	Unit consistency. Standardise or explicitly reject incompatible quantity units at item/site level. Rationale: Projected balance requires comparable units.	Must P1	All accepted calculations use one documented base UOM; missing conversions stop the affected calculation.	O3 / validity	UOM test DQ-T-08.
DQ-009	Supplier history sufficiency. Apply OTIF reliability rules only when the approved minimum observation threshold is met. Rationale: Small samples produce misleading reliability.	Must P2	Below-threshold suppliers show unavailable; threshold and above calculate from the exact approved window.	O3 / explainability	History test DQ-T-09.
DQ-010	Independent outcomes. Generate and retain simulated adverse outcome labels using logic/seeds independent of score rules. Rationale: Circular labels make recall/false-positive metrics meaningless.	Must P1-P2	A design review verifies separate generators/seeds and hidden outcomes; perturbing a score threshold does not regenerate labels.	O2 / recall and false positive	Independence review DQ-T-10.
DQ-011	Reconciliation. Reconcile source, accepted, quarantined, eligible, scored, candidate, exception, and published counts by run. Rationale: Every record needs a disposition.	Must P1-P4	For every successful test run: source = accepted + rejected and eligible = scored; unexplained difference is zero.	O5, O6	Control-total test DQ-T-11.

8. Audit and governance requirements

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
GOV-001	Append-only event history. Normal UI operations shall append events and shall not edit/delete prior controlled events. Rationale: Audit evidence must survive user changes.	Must P3	No standard role can update/delete an existing event through the UI; history remains unchanged after all workflow tests.	O5 / audit completeness	Permission/history test AT-01.
GOV-002	Event content. Capture event ID, exception ID, event type, actor/system, Melbourne timestamp, before/after, reason/source, run, and correlation/approval reference. Rationale: Reconstruction requires consistent evidence fields.	Must P3	100% of controlled events pass mandatory-field validation.	O5 / audit completeness	Schema/content test AT-02.
GOV-003	Score history. Retain every recalculated score and contribution set by PO-line episode/run. Rationale: Current score must not overwrite prior rationale.	Must P2	A 10-run episode contains 10 immutable score snapshots and matching contributions.	O3, O5	History test AT-03.
GOV-004	Approval separation. Prevent the requester from self-approving a material override in the production-target design; the MVP may demonstrate this with distinct simulated roles. Rationale: Self-approval weakens control.	Must P3	Requester-role approval attempt is blocked; distinct manager role succeeds and both identities are logged.	O5 / override governance	SoD test AT-04.
GOV-005	Override transparency. Retain original value, requested value, reason, evidence, requester, decision, approver, time, and expiry. Rationale: Overrides must not erase calculated risk.	Must P3	All approved/rejected/expired test overrides retain every field and report status.	O5 / override rate	Override test AP-01.
GOV-006	Rule change control. Require documented owner, rationale, effective date, test evidence, approval, and version for rule/threshold changes. Rationale: Silent changes invalidate comparability.	Must P2	An unapproved version cannot become active; approved version activates only on its effective boundary.	O3, O5	Change-control test AT-06.
GOV-007	Evidence references. Require resolvable evidence metadata without storing real confidential documents in the portfolio. Rationale: Actions and closure need support while data remains synthetic.	Must P3	All evidence-required actions contain type, synthetic reference, source/actor, and recorded time; broken reference test is flagged.	O5	Evidence test AT-07.
GOV-008	Evidence labelling. Label targets, simulation assumptions, simulated measured results, and limitations in documents and reporting. Rationale: Portfolio credibility depends on honest claims.	Must P0-P6	No sampled output presents a proposed target as achieved; review checklist passes 100%.	All / evidence integrity	Content review GOV-T-08.
GOV-009	Retention decision. Apply a documented configurable retention policy to run, score, event, approval, and evidence metadata after the retention decision is frozen. Rationale: History and portfolio size require an explicit policy.	Should P0-P3	Configured retention leaves the complete 24-month reference period and all acceptance-run evidence available; no silent purge occurs.	O5, O6	Retention test AT-09.
GOV-010	Administrator accountability. Attribute administrative configuration and data correction actions and report their effect. Rationale: Admins can bypass UI controls in real systems.	Must P1-P4	Every configuration/data-correction fixture records administrator, reason, time, affected objects, and before/after/checksum.	O5	Admin audit AT-10.

9. Security and access assumptions

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
SEC-001	Synthetic data only. Use no real employer, customer, supplier, or personal information in the repository or demonstration. Rationale: The public portfolio has no authority to process real data.	Must P0-P6	Repository and export scan finds no real identifiers/secrets; all entities are documented synthetic.	O5 / evidence integrity	Data scan SEC-T-01.
SEC-002	Role concept. Define permissions for buyer, planner, manager, administrator, audit reviewer, and read-only portfolio reviewer. Rationale: Workflow authority differs by role.	Must P3	A permission matrix exists and positive/negative tests pass for every controlled action.	O4, O5	Role test SEC-T-02.
SEC-003	Simulated role switching. Label MVP role switching as a simulation and never as production authentication. Rationale: The MVP cannot substantiate enterprise identity controls.	Must P3	Every role-switch screen/demo note displays the limitation; no production-security claim appears.	O5 / evidence integrity	Content review SEC-T-03.
SEC-004	Secrets management. Keep credentials, tokens, and connection secrets outside source control and public outputs. Rationale: Repository exposure is avoidable.	Must P1-P6	Secret scan is clean; configuration uses non-committed secret mechanism and sample placeholders.	O5	Secret scan SEC-T-04.
SEC-005	Database access boundary. Use separate application/reporting/admin database roles in the target PostgreSQL design; MVP permissions shall approximate least privilege where practical. Rationale: The deployed database is the control boundary.	Should P3-P4	App role cannot alter audit history/schema; reporting role is read-only; admin actions are attributable.	O5, O6	Database-role test SEC-T-05.
SEC-006	Hosted demonstration gate. Do not host until local acceptance, synthetic-data scan, secret scan, and documented limitations pass. Rationale: Public hosting expands exposure.	Must Future	No hosted endpoint is enabled without signed gate evidence; if never hosted, requirement is satisfied by documented deferral.	All	Release gate SEC-T-06.

10. Reporting requirements

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
REP-001	Operational-control page. Report current backlog by severity/status/site/owner with unassigned and overdue emphasis. Rationale: Managers need an actionable control view.	Must P4	Visual totals equal the current successful run for all filter combinations in the reconciliation suite.	O4, O6 / backlog	BI test RT-01.
REP-002	Ageing and SLA. Report ageing bands, assignment/action breaches, next-action overdue rate, and cycle time by severity. Rationale: Delay in handling is a control signal.	Must P4	All KPI values match independently calculated fixtures and defined denominators.	O4, O6 / SLA KPIs	BI test RT-02.
REP-003	Risk reasons. Report score bands, triggered reasons, contribution mix, and rule version without interpreting score as probability. Rationale: Management needs transparent drivers.	Must P4	Reason counts and points reconcile; no label uses probability/likelihood percentage language for the score.	O3, O6 / explainability	BI test RT-03.
REP-004	Supplier patterns. Report exceptions, OTIF, repeat-risk, root cause, recovery commitment, and outcome by supplier/item/site where history is sufficient. Rationale: Recurring patterns should be separated from noise.	Must P4	Insufficient-history suppliers are marked; fixture aggregations match expected results.	O6 / supplier insight	BI test RT-04.
REP-005	Interventions and overrides. Report action types, mitigations, approved/rejected/expired overrides, accepted residual risk, and later outcomes. Rationale: Learning requires visibility of intervention, not just alert volume.	Must P4	Every reference action/override appears once and is linked to the correct episode/outcome.	O5, O6	BI test RT-05.
REP-006	Detection quality. Report recall, false-positive rate, useful lead time, workload volume, and missed-event review using independent simulated outcomes. Rationale: Threshold calibration needs balanced evidence.	Must P4	Metrics reproduce the approved outcome-window fixture; denominators and exclusions are displayed.	O1, O2 / recall, FP, lead time	Metric test RT-06.
REP-007	Data and run health. Report latest successful run, failed runs, source freshness, validation defects, quarantine, coverage, and batch duration. Rationale: Users must know whether data is safe to use.	Must P4	Latest-success and failed-run scenarios display correct status and age; no failed run is presented as current.	O6 / freshness, validity	BI test RT-07.
REP-008	Daily refresh. Refresh management outputs daily from curated PostgreSQL views or controlled extracts. Rationale: The charter sets daily management monitoring.	Must P4	Latest successful dataset is <=26 hours old after scheduled daily cycle; exception periods are disclosed.	O6 / freshness	Refresh test RT-08.
REP-009	Reconciliation. Reconcile operational and Power BI totals by run, date, severity, status, supplier, and site. Rationale: Contradictory interfaces undermine trust.	Must P4	Difference is zero for every mandatory reconciliation dimension in three acceptance runs.	O6 / reconciliation	Control-total test RT-09.
REP-010	Proposed-target labelling. Display proposed targets separately from measured simulated results and real-world outcomes. Rationale: Targets are hypotheses, not achievements.	Must P4	Every KPI card/page identifies target and measured status; unmet targets remain visible.	All / evidence integrity	Presentation review RT-10.

11. Exception lifecycle requirements

The lifecycle distinguishes analytical candidates from managed exceptions. Candidate risk is created by processing; Open and later states represent operational control records. “Resolved” means the risk condition has been addressed or accepted with evidence; “Closed” means an authorised reviewer has verified completion. Suppression never deletes history.

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
LIF-001	Candidate risk. Automatically create one candidate per eligible PO line/run with score and disposition. Rationale: Not every evaluated line should enter the queue.	Must P2	Candidate count equals eligible scored count and each has a disposition.	O3 / coverage	LC-01.
LIF-002	Candidate to Open. Automatically open an exception when the active opening threshold/trigger and deduplication rules are met. Rationale: Material candidates must enter operational control consistently.	Must P3	All opening boundary fixtures create exactly one Open episode; non-triggers do not.	O2, O3	LC-02.
LIF-003	Open to Assigned. Automatically assign where a valid mapping exists; otherwise retain Open in an unassigned queue until human assignment. Rationale: Automation can allocate but not conceal gaps.	Must P3	Mapped/unmapped fixtures produce expected owner/SLA state.	O4 / assignment	LC-03.
LIF-004	Assigned to Investigating. Require the owner or authorised manager to begin investigation explicitly. Rationale: Active work must be attributable.	Must P3	Transition records actor/time and rejects unauthorised user.	O4, O5	LC-04.
LIF-005	Investigating to Action Agreed. Require a human-recorded action plan, owner, next-action date, and applicable commitment/evidence. Rationale: A diagnosis without a plan is not controlled response.	Must P3	Missing plan fields block transition; valid record creates an event.	O4, O5	LC-05.
LIF-006	Action Agreed to Monitoring. Require a human confirmation that the agreed action was initiated and define the monitored commitment/date. Rationale: Monitoring needs a known expected outcome.	Must P3	Transition rejects missing monitored date/condition and records initiator.	O5	LC-06.
LIF-007	Monitoring to Resolved. Allow automatic resolution only for objective source events approved in the rule catalogue (for example full receipt); require human resolution for substitution, demand change, cancellation, or accepted residual risk. Rationale: Objective evidence can reduce effort; judgement remains accountable.	Must P3	Each objective and judgement route follows its configured authority/evidence path and assigns a resolution code.	O5 / cycle time	LC-07.
LIF-008	Resolved to Closed. Require authorised human verification of resolution evidence before Closed. Rationale: Closed is a control sign-off, not merely a low score.	Must P3	No automatic Closed transition occurs; verifier is distinct where approval policy requires.	O5 / audit	LC-08.
LIF-009	Suppressed. Allow automatic suppression for exact duplicate/cooldown rules and human suppression only with reason, expiry, and manager approval for material risk. Rationale: Suppression manages noise but can hide risk.	Must P3	Suppressed records retain score/history, reason, authority, start/expiry, and remain reportable.	O1, O5	LC-09.

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
LIF-010	Reopened. Automatically reopen on configured material deterioration/objective recurrence and permit authorised human reopen with reason. Rationale: Risk may recur after mitigation or closure.	Must P3	Reopen creates one linked event, preserves prior history, and resets applicable SLA.	O2, O5	LC-10.
LIF-011	Transition rejection. Reject all transitions not included in the approved state matrix. Rationale: State integrity must be enforceable.	Must P3	Negative test covers every unapproved from/to pair with zero state mutation.	O5	LC-11.
LIF-012	Lifecycle timestamps. Record entered/exited timestamps and responsible actor/system for every status. Rationale: Ageing, SLA, and reconstruction require event time.	Must P3	100% of status episodes have internally consistent times and event links.	O4-O6	LC-12.

Lifecycle transition matrix

From	To	Decision mode	Trigger	Control note
No record	Candidate risk	Automatic	Successful eligibility and scoring	One per PO line/run; analytical record.
Candidate risk	Open exception	Automatic	Opening rule met; no active duplicate	Starts operational episode and SLA.
Candidate risk	Suppressed	Automatic	Exact duplicate/cooldown/noise rule	Retained and reportable; no deletion.
Open exception	Assigned	Automatic or human	Valid mapping or authorised assignment	Unmapped stays Open/unassigned.
Assigned	Investigating	Human	Owner begins work	Actor/time required.
Investigating	Action agreed	Human	Documented plan and due date	Commitment/evidence by action type.
Action agreed	Monitoring	Human	Action initiated; monitoring condition set	Next review/commitment required.
Monitoring	Resolved	Automatic or human	Objective source evidence, or authorised judgement with evidence	Automatic limited to approved objective events.
Resolved	Closed	Human only	Authorised verification	Never AI or score-only.
Any active	Suppressed	Automatic or approved human	Duplicate/cooldown or approved suppression	Material manual suppression needs manager and expiry.
Resolved/Closed/Suppressed	Reopened	Automatic or human	Material deterioration/recurrence or authorised reason	Prior history preserved; SLA reset.

12. Ownership, SLA, escalation, and approval requirements

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
SLA-001	Default ownership. Assign by approved buyer-group mapping with site/supplier exceptions and audited manager reassignment. Rationale: A deterministic default limits unowned work.	Must P3	Mapping fixture assigns 100% expected owners; exceptions are unassigned and visible.	O4 / assignment	OW-01.
SLA-002	High/critical assignment. Assign >=95% of high/critical exceptions within 4 Melbourne business hours of Open. Rationale: This is the charter ownership target.	Must P3	30-run SLA report meets >=95% or shortfall is disclosed with breached records. Proposed target.	O4 / assignment SLA	OW-02.
SLA-003	Medium assignment. Assign Medium exceptions within 1 Melbourne business day. Rationale: The initial response matrix sets this expectation.	Should P3	All reference Medium cases calculate the correct due time; breach state appears after due. Proposed target.	O4	OW-03.
SLA-004	Action plan. Require Critical action plan within 4 business hours and High within 8 business hours. Rationale: Higher consequence requires faster response.	Must P3	Boundary/DST/holiday fixtures calculate due time and report breach correctly. Proposed target.	O4 / action SLA	OW-04.
SLA-005	Business calendar. Calculate SLAs using configurable Australia/Melbourne business hours, weekends, holidays, and pause rules. Rationale: Elapsed clock time is not business response time.	Must P3	Approved calendar cases, including DST and holiday, match expected minutes exactly.	O4	OW-05.
SLA-006	Escalation. Expose and escalate unassigned/overdue High/Critical exceptions to the manager queue without changing owner or status silently. Rationale: Breaches need visible intervention.	Must P3	Breach scenario appears in manager queue at due time and creates an escalation event.	O4, O5	OW-06.
SLA-007	Reassignment. Permit authorised reassignment with reason while preserving prior owner and SLA history; define whether SLA resets in the frozen policy. Rationale: Reassignment can otherwise game performance.	Must P3	Reassignment event contains before/after/reason; SLA behavior matches the approved reset policy.	O4, O5	OW-07.
SLA-008	Material approvals. Require manager approval for severity reduction, criticality override, accepted residual risk, manual material suppression, and closure without objective receipt/substitution evidence. Rationale: These decisions reduce visible or accepted exposure.	Must P3	All five routes block without approval and succeed with complete distinct-role approval.	O5	AP-01.
SLA-009	Approval decision. Record approve/reject, rationale, conditions, expiry, approver, and time; rejection shall not apply requested change. Rationale: Approval is a decision record.	Must P3	Approved, conditional, expired, and rejected fixtures produce exact expected state.	O5	AP-02.

13. AI governance requirements for the optional phase

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
AI-001	Optional and gated. Begin AI only after P1-P4 data, risk, workflow, audit, tests, and BI exit criteria pass. Rationale: AI must not distract from the controlled core.	Must P6	AI feature flag remains off until signed P4 gate evidence exists.	O7	AI gate AI-T-01.
AI-002	Permitted uses. Limit AI to context summary, 2-3 constrained next-action options, and communication draft. Rationale: These uses assist judgement without owning it.	Must P6	Output schema accepts only the three permitted types; unsupported request is refused.	O7	Schema test AI-T-02.
AI-003	Prohibited mutations. Prevent AI from creating, closing, reprioritising, reassigning, approving, suppressing, resolving, or executing operational actions. Rationale: Human accountability is binding.	Must P6	Permission/adversarial suite produces zero operational mutations across all prohibited verbs.	O7 / 0 autonomous changes	AI-T-03.
AI-004	Human review. Require a human to review, edit if needed, approve or reject every AI output before operational use. Rationale: Plausibility is not authority.	Must P6	100% of generated outputs have a review decision; unreviewed output cannot be copied to final action/communication state.	O7 / review rate	AI-T-04.
AI-005	Approved data boundary. Send only synthetic, minimised, approved fields to the model. Rationale: Data minimisation reduces exposure and irrelevant context.	Must P6	Payload inspection shows only whitelisted fields and no secrets/raw evidence documents.	O7	Payload test AI-T-05.
AI-006	AI audit record. Store use case, input reference/hash, approved context fields, prompt/template version, model/version, output, reviewer, decision, edits, time, and final linked action. Rationale: Evaluation and challenge require full lineage.	Must P6	100% of AI test outputs contain all audit fields and link to one exception.	O7	Audit test AI-T-06.
AI-007	Structured uncertainty. Require output to identify missing context and avoid presenting the risk priority score as probability. Rationale: AI can amplify false precision.	Must P6	Adversarial cases with missing data produce an uncertainty/missing-context field and no probability interpretation.	O3, O7	Content test AI-T-07.
AI-008	Kill switch. Provide a documented, tested control that disables AI without affecting scoring, queue, lifecycle, reporting, or audit. Rationale: Core operations must not depend on AI.	Must P6	With AI disabled, all P1-P4 regression tests pass and AI calls are zero.	O7	Kill-switch test AI-T-08.
AI-009	Evaluation. Evaluate groundedness, permitted-action compliance, harmful omission, consistency, and reviewer acceptance on a frozen scenario set. Rationale: A demo requires evidence beyond one good example.	Should P6	Evaluation report covers >=30 cases, includes failures, and meets 100% prohibited-action compliance; other thresholds are frozen before build.	O7	Evaluation AI-T-09.

14. MVP requirements

The MVP is the smallest complete operational control slice. Phase 4 is the core portfolio finish line. P5 automation and P6 AI are not MVP prerequisites.

MVP gate	Included requirement IDs	Exit evidence
P0 - approved design	BR-001 to BR-008; decisions DL-001 to DL-018 frozen or explicitly accepted as risks	Approved BRS, decision log, process/state model, test plan.
P1 - data foundation	FR-001 to FR-005; DQ-001 to DQ-011; NFR-003, NFR-006, NFR-007, NFR-009; SEC-001, SEC-004	Reproducible 24-month dataset; validation/quarantine/lineage and independent outcomes pass.
P2 - risk engine	FR-006 to FR-010; GOV-003, GOV-006; NFR-001, NFR-008; RULE-001 to RULE-014	100% explainable deterministic scores; workload/detection evidence or disclosed shortfall.
P3 - operational workflow	FR-011 to FR-024, FR-027; lifecycle, ownership/SLA, governance, and role requirements	End-to-end golden episode; no destructive UI history; approval and negative-transition tests pass.
P4 - management reporting and local release	FR-025, FR-028; REP-001 to REP-010; NFR-002, NFR-004; SEC-005	Operational/BI totals reconcile; freshness/failure visible; local demo accepted.

Prioritised MVP sequence

1. Freeze modelling decisions and acceptance fixtures.
2. Prove data quality, lineage, and independently generated outcomes.
3. Prove deterministic, explainable scoring and queue-volume calibration.
4. Prove one complete exception episode including assignment, action, approval, resolution, closure, suppression, and reopening.
5. Prove audit completeness and negative permissions/transitions.
6. Reconcile the operational store and Power BI.
7. Package the local demonstration and evidence; only then consider scheduling, notifications, hosting, or AI.

15. Deferred and future requirements

Timing / priority	Reference	Boundary
P5 optional	FR-026	GitHub Actions scheduled processing/tests/export after local core acceptance.
P5 optional	Notification design only	Email or one simulated channel after queue/SLA/audit acceptance; not a substitute for queue.
P6 optional	AI-001 to AI-009	Controlled summaries/options/drafts only.
Future	SEC-006	Hosted demonstration after local, synthetic-data, secret, and limitation gates.

Timing / priority	Reference	Boundary
Future	Production integration	ERP/EDI/WMS/supplier portals/real-time events.
Future	Enterprise identity	SSO, production authentication, mature RBAC/SoD, penetration testing.
Future	Advanced planning	Probabilistic inventory, optimisation, production scheduling, routing, sourcing.
Will Not - current release	Autonomous execution	Order changes, cancellation, expedite purchase, premium freight approval, supplier sanctions, inventory allocation, or AI workflow mutation.
Will Not - evidence	Realised-benefit claims	No claim of actual cost, service, or labour savings from synthetic tests.

16. Business rules

The initial scorecard below is a proposed portfolio starting policy, not a calibrated production policy. Rules and thresholds must be versioned and may be changed only through GOV-006. “Risk priority score” is an ordinal prioritisation index, not a probability.

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
RULE-001	Eligibility. Score a PO line only when it is open under approved status/quantity rules and has not been fully received/cancelled. Rationale: Defines the controlled population.	Must P2	Eligibility fixture classifies every case correctly.	O3 / coverage	RE-01.
RULE-002	Score range. Sum approved contributions and mitigations, floor at 0 and cap at 100. Rationale: Stable scale supports severity bands.	Must P2	All extreme fixtures return 0-100 and reproduce contribution math.	O3	RE-02.
RULE-003	Overdue signal. Proposed R01: add 30 points when past latest accepted confirmed delivery date and quantity remains unreceived. Rationale: Delay is already materialising.	Must P2	Boundary-date/partial-receipt fixtures match rule. Proposed starting rule.	O2, O3	RE-03.
RULE-004	Stockout exposure. Proposed R02: add 25 points when deterministic projected available balance falls below the approved threshold before expected receipt. Rationale: Direct service/production consequence.	Must P2	Projection fixture matches hand calculation. Proposed starting rule.	O2, O3	RE-04.
RULE-005	Missing confirmation. Proposed R03: add 15 points when need date is within 7 calendar days and no valid confirmation/ASN exists. Rationale: Low visibility near response deadline.	Must P2	Day-7/day-8 and confirmation-validity boundaries match. Proposed starting rule.	O2, O3	RE-05.
RULE-006	Supplier reliability. Proposed R04: add 15 points when trailing OTIF <85% and minimum history is met. Rationale: Persistent supplier deterioration matters.	Must P2	OTIF and minimum-history boundaries match. Proposed starting rule.	O2, O3	RE-06.
RULE-007	Lead-time variance. Proposed R05: add 10 points when expected lead time exceeds item-supplier baseline by >20%. Rationale: Unusual delay is an early signal.	Should P2	20% and >20% fixtures match; insufficient baseline is unavailable. Proposed starting rule.	O2, O3	RE-07.

ID	Requirement statement and business rationale	Priority / phase	Measurable acceptance criterion	Charter objective / KPI	Proposed test method
RULE-008	Item criticality. Proposed R06: add 5 points for approved operational-criticality flag. Rationale: Criticality raises consequence, not likelihood.	Must P2	Flagged case adds exactly 5 and display identifies consequence. Proposed starting rule.	O3	RE-08.
RULE-009	Documented mitigation. Proposed M01: subtract 15 when approved substitute or protected stock covers the need. Rationale: Verified mitigation reduces consequence.	Must P2-P3	Mitigation applies only with coverage calculation and evidence/approval; expiry reverses it prospectively.	O3, O5	RE-09.
RULE-010	Severity. 0-29 Monitor; 30-49 Medium; 50-69 High; 70-100 Critical. Rationale: Initial response matrix needs deterministic bands.	Must P2	All boundaries map exactly. Proposed starting thresholds.	O3	RE-10.
RULE-011	No probability claim. Never interpret score x as x% chance of delay/stockout. Rationale: The engine is rule-based, not calibrated probability.	Must P2-P6	No UI/report/document/AI output equates score to probability.	O3, O7	Content scan RE-11.
RULE-012	Correlated-signal control. Document dependencies, test contribution correlation, and cap overlapping categories where approved. Rationale: Overlapping signals can inflate severity.	Must P2	Calibration report includes correlation/contribution analysis and configured caps; no unexplained double-counting.	O1-O3 / FP rate	Calibration RE-12.
RULE-013	Opening and suppression. Opening threshold, cooldown, duplicate key, and material-deterioration threshold shall be frozen before workflow build. Rationale: Lifecycle volume depends on these rules.	Must P0-P3	Approved decision log values exist and boundary fixtures pass.	O1, O2, O4	Design gate RE-13.
RULE-014	Date and partial receipt hierarchy. Use the frozen hierarchy among need date, latest accepted promise, expected receipt, and receipt; evaluate only unreceived quantity under documented partial-receipt logic. Rationale: Ambiguous dates/quantities change delay and exposure.	Must P0-P2	Hand-calculated multi-date/partial receipt fixtures match 100%.	O2, O3	Rule test RE-14.

17. Constraints and dependencies

Constraint / dependency	Implication	Required treatment
Single developer	One analyst must design, build, test, document, and demonstrate.	Keep P5/P6 optional; prefer a complete vertical slice.
Synthetic evidence	No real users, source systems, communications, or cost outcomes.	Treat KPIs as proposed acceptance targets and simulated measured results.
PostgreSQL primary	Adds configuration, migration, backup, and deployment work.	Freeze schema semantics early; keep SQLite tests subordinate.
Streamlit boundary	Suitable for workflow demonstration, not proof of enterprise case management/security.	Label simulated roles; document target controls.
Power BI environment	PBIX sharing/refresh can depend on licences, gateway, and hosting.	Use controlled extracts or PostgreSQL views; local demonstration first.
GitHub Actions	Hosted runners are ephemeral and schedules are not enterprise-grade.	Use for reproducible processing/tests/export, not operational state.
Australia/Melbourne	DST, business hours, and holiday calendar affect SLA.	Freeze calendar and test DST boundaries.
Outcome validity	Score and simulated labels can become circular.	Independent seeds/logic and unseen scenarios are mandatory.
External supplier action	System records contact/commitment but cannot verify email/call content.	Use structured evidence references; integration remains out of scope.

18. Requirements traceability matrix

The matrix groups related requirement IDs where they support the same objective/capability/KPI. Detailed acceptance and test evidence remain in the requirement tables.

Business objective	Requirement(s)	System capability	KPI / proposed target	Proposed test evidence
O1 Focus effort	BR-001; US-001/002; FR-012-016; RULE-012/013	C04/C05 - lifecycle and queue	Screening <=35 min; queue volume; FP <=35%	UT-01 timed script; ordering/filter fixtures; calibration report.
O2 Detect earlier	BR-002; US-003; FR-006-010/023; DQ-004/005/007/010; RULE-003-007/014	C01/C03/C04 - scenarios, score, reopen	Recall >=70%; lead time >=2 days	Unseen-scenario backtest AT-02; missed-event review.
O3 Standardise priority	BR-003; FR-003-010; NFR-001/003/008/009; DQ-001-011; GOV-003/006; RULE-001-014	C02/C03 - validation and explainable engine	Coverage >=99.5%; explainability 100%; batch <=5 min	DQ suite; rule boundary suite; determinism/performance tests.
O4 Create ownership	BR-004; US-001/004; FR-013-019; LIF-002-006/010; SLA-001-007	C05/C06 - queue, ownership and SLA	Assignment >=95% in 4h; action >=90%; overdue <10%	Ownership/calendar/escalation scenarios; timed queue test.
O5 Preserve evidence	BR-005; US-002/004-006; FR-002/011-024; GOV-001-010; SEC-002/005; LIF-001-012; SLA-008/009	C07/C08/C09 - action, approval, resolution	Audit completeness 100%	Golden episode; negative state/role tests; event completeness.

18. Requirements traceability matrix (continued)

Business objective	Requirement(s)	System capability	KPI / proposed target	Proposed test evidence
O6 Management learning	BR-006; US-007; FR-025/026; REP-001-010; NFR-004/005; DQ-011	C10/C11 - BI and scheduling	Freshness <=26h; batch >=98%; reconciliation zero	BI control totals; failed-run/freshness scenario; 30-run report.
O7 Govern AI	BR-007; AI-001-009; SEC-001/004; RULE-011	C12 - controlled AI support	100% reviewed; 0 autonomous mutations	AI permission/adversarial/evaluation/kill-switch evidence.
All / evidence integrity	BR-008; GOV-008; SEC-003/006; REP-010; FR-027/028	Governance and release gates	No unsupported achieved-benefit claim	Document/content scan; release checklist; limitation review.

19. Acceptance approach

Acceptance is evidence-based and phased. A successful technical run is necessary but not sufficient: control behavior, negative cases, reconciliation, documentation, and honest limitations must also pass. Proposed targets that are missed shall be recorded as failed or partially met, not re-labelled after the test.

Gate	Required evidence	Method	Acceptance authority
P0 Design acceptance	Approved BRS, process/state model, decision log, rule catalogue, data dictionary outline, test fixtures, and scope gates.	Requirements review; ambiguity log; traceability inspection.	Business analyst / project owner.
P1 Data acceptance	Reproducible baseline volumes; independent outcomes; critical validation gate; quarantine; lineage; control totals.	DQ positive/negative/boundary tests and reproducibility hashes.	Data/system administrator; reviewer.
P2 Risk acceptance	Deterministic score; rule version; contribution explanation; performance; recall/FP/lead-time/workload evidence.	Hand calculations, boundary tests, unseen-scenario backtest, sensitivity analysis.	Supply-chain manager as simulated rule owner.
P3 Workflow acceptance	Golden end-to-end episode; state controls; ownership/SLA; actions; approvals; suppression; reopen; audit.	Given/When/Then scripts, negative transitions, role matrix, audit completeness.	Buyer/planner/manager/audit reviewer roles.
P4 BI/local release	Power BI reconciliation/freshness/failure; local deployment; documentation and limitations.	Control totals, refresh scenario, timed usability, demo rehearsal.	Manager and independent portfolio reviewer.
P5 optional	Scheduled repeatability and visible failure; persistence boundary.	30-run schedule report and failure injection.	Administrator.
P6 optional	Advisory-only AI, human review, audit, evaluation, kill switch.	Permission/adversarial tests and >=30-case evaluation.	Governance reviewer and manager.

Proposed acceptance evidence register

AT-02 unseen-scenario detection report; AT-05 golden audit episode; DQ-T-01 to DQ-T-11 data-quality pack; RE-01 to RE-14 deterministic rule pack; LC-01 to LC-12 lifecycle pack; OW/AP ownership and approval pack; PT performance/repeatability/atomicity report; RT-01 to RT-10 Power BI reconciliation pack; UT-01 timed screening protocol; OPS-T-01 30-run automation evidence; AI-T-01 to AI-T-09 optional AI governance pack.

20. Open questions and design risks

ID	Question / risk	Decision required	Exposure	Timing
Q01	Operational consequence	How will customer-service-critical and production-critical items differ in consequence, escalation, and evidence?	High	Freeze before item/site model.
Q02	Episode key	What exact combination identifies one PO-line risk episode across split schedules, amendments, and partial receipts?	High	Freeze before keys/constraints.
Q03	Date hierarchy	Which date is authoritative when requested need, confirmed promise, expected receipt, and revised supplier date disagree?	High	Freeze before risk facts.
Q04	Partial receipts	How are residual quantity, over-receipt tolerance, late partials, and multiple receipt dates evaluated?	High	Freeze before eligibility.
Q05	Projected balance	Exact forecast/requirements source, horizon, safety stock, allocations, substitutes, and missing-demand behavior?	High	Freeze before stockout rule.
Q06	Criticality	Who owns criticality, what is its effective date/site scope, and what approval/expiry governs overrides?	High	Freeze master-data governance.
Q07	Queue capacity	Is 15-25 new exceptions/day per buyer the correct calibration assumption, and what miss tolerance applies to critical items?	High	Freeze before thresholds.
Q08	Supplier OTIF	What observation window, late tolerance, minimum sample, cancellations, and partial-delivery treatment apply?	High	Freeze before supplier metric.
Q09	Suppression/reopen	What cooldown, material-deterioration threshold, suppression expiry, and reopen SLA reset apply?	High	Freeze before lifecycle build.
Q10	Approval authority	Which roles/thresholds approve each severity reduction, residual risk, suppression, criticality change, and exceptional closure?	High	Freeze permission matrix.
Q11	Business calendar	What operating hours, holiday source, after-hours handling, and SLA pause rules apply?	Medium	Freeze before SLA fields.
Q12	Retention	How long are score, event, approval, evidence metadata, quarantine, and runs retained?	Medium	Freeze before physical design.
Q13	Outcome window	What future window defines an adverse outcome and false positive by risk type?	High	Freeze before backtest.
Q14	Power BI path	Direct PostgreSQL views or curated CSV/Parquet for final local demonstration and portable evidence?	Medium	Freeze before semantic design.
Q15	Performance environment	What hardware and cold/warm-run protocol define the <=5-minute target?	Medium	Freeze before performance evidence.
Q16	Concurrency	How many simultaneous demo users/writers must PostgreSQL and Streamlit support?	Medium	May be a documented MVP limit.
Q17	Evidence format	Which synthetic evidence types/references are mandatory for each action/resolution?	Medium	Freeze before workflow forms.
Q18	Public package	Which sample data, PBIX, screenshots, exports, and logs may be public?	Medium	Freeze before portfolio release.

Appendix A. Prioritised MVP requirement list

Priority order	Requirement group	Why it precedes the next group
1	P0 decisions; BR-001 to BR-008; RULE-013/014	Keys, dates, lifecycle and thresholds determine every downstream table and test.
2	FR-001 to FR-005; DQ-001 to DQ-011; SEC-001/004	Bad or circular data invalidates risk metrics.
3	FR-006 to FR-010; RULE-001 to RULE-012; GOV-003/006; NFR-001/003/008	Score must be deterministic, explainable, versioned, and workload-calibrated.
4	FR-011 to FR-024; LIF-001 to LIF-012; SLA-001 to SLA-009; GOV-001/002/004/005/007/010; SEC-002/003	The operational control loop is the central portfolio evidence.
5	REP-001 to REP-010; FR-025/028; NFR-002/004/006/007/009; SEC-005	Reporting follows stable workflow semantics and must reconcile.
6	FR-026/027, SEC-006, AI-001 to AI-009	Automation, notifications, hosting, and AI add value only after the core is accepted.

Appendix B. Requirements too ambitious for one developer

Ambitious requirement	Why it exceeds a solo portfolio build	Recommended treatment
Production-grade multi-user security	SSO, identity lifecycle, enterprise RBAC/SoD, penetration testing, secrets rotation, 24/7 monitoring.	Document target architecture; do not claim production security.
Tamper-proof audit and disaster recovery	Immutable/WORM storage, independent admin controls, backup restore drills, legal retention.	Demonstrate append-only UI/event patterns and disclose admin limits.
Live ERP/EDI/WMS integration	Source contracts, change data capture, error recovery, reconciliation, supplier master ownership.	Use synthetic extracts and interface specifications only.
Robust notification ecosystem	Email/Teams/Slack delivery, retries, bounce handling, preferences, throttling, on-call ownership.	In-app queue first; at most one simulated channel later.
Hosted production deployment	Network, certificates, domains, secure secrets, durable scheduling, backup, monitoring, cost management.	Local demonstration first; hosting optional.
Advanced inventory prediction/optimisation	Forecast uncertainty, allocation, safety stock, substitutions, production constraints, probabilistic receipt.	Use deterministic projected balance and state limitations.
Full AI assurance program	Model risk assessment, privacy/legal review, red teaming, continuous monitoring, vendor controls.	Optional constrained proof of concept only after core acceptance.
Broad practitioner validation	Representative sampling, ethics/privacy, structured usability research, multiple sites/roles.	Seek 1-3 practitioner reviews and label as design feedback, not endorsement.
Enterprise Power BI service operation	Gateway, scheduled cloud refresh, RLS, deployment pipelines, licences, tenant governance.	Deliver local PBIX/model evidence and documented target pattern.

Appendix C. Charter conflicts, ambiguities, and inconsistencies

Issue	Conflict / ambiguity	BRS treatment
Document status	Charter cover says “Proposed charter”; commissioning instruction calls it approved Version 1.0.	Treat charter as approved governing source for this BRS; retain original status discrepancy in document control.
Primary database	Charter core scope recommends SQLite local MVP and PostgreSQL later; confirmed baseline makes PostgreSQL primary deployed DB and SQLite development/testing only.	Confirmed baseline supersedes charter recommendation.
Lifecycle terminology	Charter recommends New/Assigned/Investigating/Action Agreed/Monitoring/Resolved/Closed; confirmed baseline requires candidate risk, open exception, assigned, investigating, action agreed, monitoring, resolved, closed, suppressed, reopened.	Use the ten-state model in this BRS; map “New” to Open exception.
Automatic resolution vs human evidence	Charter says engine may resolve candidates/exceptions but closure requires evidence and human control.	Limit automatic Resolved to approved objective source events; Closed always requires human verification.
AI approval wording	Charter O7 says human approval for recommendations that could affect operations; confirmed baseline prohibits AI from approving or executing any decision.	Require review of every AI output and zero AI mutations.
Primary grain wording	Charter says one open PO line at one run, while one persistent exception spans runs; baseline says primary operational grain is one PO-line risk episode.	Use episode as operational grain; candidate score is PO-line/run analytical grain.
SLA detail	Charter KPI says high/critical action within 8 hours; severity table says Critical action plan within 4 hours.	Use Critical 4 hours, High 8 hours, both proposed targets.
Metric wording	Charter calls risk a 0-100 score and later warns against probability interpretation.	Use “risk priority score” everywhere and explicitly prohibit probability claims.
Notifications/deployment	Charter makes them optional but not fully sequenced.	Confirmed baseline freezes queue before email and local before hosted.
Auditability claim	Charter acknowledges append-only UI does not guarantee database immutability.	Specify demonstrated UI/event controls and avoid claiming tamper-proof audit.

Appendix D. Critical self-review

Assessment	Critical analysis	Required response
Strength	Traceability is unusually strong for a portfolio specification: objectives, requirements, capabilities, KPIs, tests, and phased gates are connected.	Preserve the IDs and evidence register through implementation.
Strength	The lifecycle separates candidate analysis, operational work, resolution, closure, suppression, and reopening, and states who/what may transition.	Validate it against practitioner feedback before freezing.
Strength	The specification avoids probability language and autonomous-AI claims.	Run a terminology scan across UI, BI, README, and demo.

Assessment	Critical analysis	Required response
Weakness	Many thresholds remain proposed assumptions without real buyer baseline or cost/service evidence.	Do not optimise to synthetic metrics alone; disclose sensitivity and missed cases.
Weakness	The specification is broad for one developer, even with optional phases.	Stop at P4 if schedule slips; do not cut audit, DQ, or reconciliation first.
Weakness	Automatic resolution for objective events still needs careful handling of late corrections, reversals, and partial receipts.	Freeze reversal/reopen logic and test corrected receipts.
Weakness	A role simulation cannot prove authentication, segregation, or administrator resistance.	Label the boundary on screens and in the demo; test only demonstrable permissions.
Weakness	The 0-100 score can create false precision and correlated rules may dominate.	Show contributions/bands, cap overlaps, and report sensitivity/workload.
Weakness	Power BI reconciliation can be correct while operational definitions are wrong.	Have a reviewer challenge KPI definitions and denominators, not just totals.
Weakness	The BRS is detailed enough to encourage premature design lock-in.	Treat open decisions as gates; version changes rather than forcing ambiguous requirements into the schema.
Overall judgement	Proceed with conditions: complete P0-P4 as one tested vertical slice; keep P5/P6 and hosted features optional.	Judge success by evidence and control completeness, not technology count.

Appendix E. Recommended pre-modelling decision log

ID	Decision	Decision to freeze	Owner role	Latest point
DL-001	Business scenario consequence	Define service-critical versus production-critical outcomes by site/item.	Business owner	Before entities
DL-002	Episode and deduplication key	Freeze PO-line episode identity, split schedules, amendments, and recurrence link.	BA + data owner	Before keys
DL-003	Date hierarchy	Freeze requested need, accepted promise, expected receipt, revised date, and receipt precedence.	Procurement manager	Before risk facts
DL-004	Partial receipt logic	Freeze residual quantity, tolerances, multiple receipts, and corrected/reversed receipts.	Procurement + warehouse	Before eligibility
DL-005	Projected balance	Freeze demand source, horizon, safety stock, allocations, substitutions, and missingness.	Inventory planner	Before stock logic
DL-006	Criticality governance	Freeze owner, site scope, effective dates, override authority, and expiry.	Supply-chain manager	Before item master
DL-007	Supplier-performance definition	Freeze OTIF window, tolerance, minimum history, cancellation/partial rules.	Procurement manager	Before supplier metric
DL-008	Risk rule version 1	Approve signals, points, caps, bands, opening threshold, and unavailable treatment.	Supply-chain manager	Before score model
DL-009	Outcome/evaluation policy	Freeze adverse outcomes, windows, hidden labels, scenario splits, recall/FP denominators.	BA + reviewer	Before backtest data
DL-010	Queue workload assumption	Freeze buyer capacity, critical miss tolerance, and calibration objective.	Supply-chain manager	Before thresholds

ID	Decision	Decision to freeze	Owner role	Latest point
DL-011	Ownership mapping	Freeze buyer group/site/supplier precedence and unassigned handling.	Procurement manager	Before owner fields
DL-012	Lifecycle policy	Approve state names, allowed transitions, auto/human modes, suppression/cooldown/reopen.	Control owner	Before event model
DL-013	SLA calendar	Freeze business hours, holidays, DST, pauses, reassignment/reopen reset.	Control owner	Before SLA timestamps
DL-014	Approval matrix	Freeze material actions, approver roles, self-approval rule, conditions, expiry.	Governance owner	Before role/event model
DL-015	Resolution evidence	Freeze codes and mandatory evidence by receipt, substitute, demand, cancellation, residual risk.	Control owner	Before resolution model
DL-016	Retention and correction	Freeze retention, purge/archive, data corrections, and admin audit.	Data/governance owner	Before physical model
DL-017	BI publication path	Freeze PostgreSQL views versus controlled extracts, current-run pointer, reconciliation dimensions.	Data/BI owner	Before analytics model
DL-018	Deployment/public evidence	Freeze local environment, performance protocol, public files, hosting gate, and secrets boundary.	Project owner	Before release design

RECOMMENDED APPROVAL CONDITION Approve this BRS only after DL-002 through DL-015 are either frozen or explicitly accepted as dated assumptions. Data modelling before those decisions would embed avoidable ambiguity in keys, dates, lifecycle, approvals, and outcome measures.